

The Effects of *Bifidobacterium longum* on the Immune System

Abstract

Bifidobacterium longum is a species of bacteria that is one of the most abundant microbes present in an individual's gut microbiome at the beginning of their life. The presence of *B. longum*, in addition to its four subspecies, have shown to improve the functioning of the immune system. They do this by producing essential metabolites that assist the immune system such as acetic acid and indole-3-lactic acid. Despite its beneficial impacts, the population size of *B. longum* tends to decrease with age, changing the gut microbiome. This can be inconvenient as lower population sizes have been associated with the development of autoimmune diseases and disorders. For instance, lower population sizes of *B. longum* have been observed in individuals who develop celiac disease, inflammatory bowel disease, and irritable bowel syndrome. As a result, individuals are more vulnerable to pathogens as their immune system weakens. This impairment can lead to uncontrolled immune system responses in addition to unchecked inflammation, potentially causing damage to the host. Current studies show promising results for reintroducing *B. longum* into the microbiome, ultimately providing relief for these individuals. As an example, *B. longum* supplementation has shown to regulate the immune system by increasing cytokines that are anti-inflammatory while decreasing those that are pro-inflammatory. Along with influencing the production of cytokines, the presence of *B. longum* can influence the activity of other cells such as natural killer cells or Paneth cells. While these studies show promising results for using *B. longum* to regulate the immune system, much remains unknown about the exact mechanisms behind these improvements.

Introduction

The gut microbiome is a dynamic environment consisting of many microbes interacting with one another. The presence and activity of these microbes have the ability to influence the host. One common microbe that can be found in the gut microbiome is *Bifidobacterium longum*. Colonizing the gut during infancy and remaining well into advanced age, *B. longum* is one of the most prevalent microbes found in the gut microbiome of healthy individuals (1). However, as an individual begins to age or develop autoimmune diseases, the population size of all *B. longum* subspecies begins to dwindle and the strength of the individual's immune system begins to wane (2, 3). For instance, with aging comes constant, unchecked inflammation and increased vulnerability to disease. In the case of autoimmune diseases, perhaps an individual has developed celiac disease and is newly sensitive to gluten due to imbalances in the gut microbiome. In both of these instances, the composition of the gut microbiome fluctuates and the immune system is altered. Current studies suggest that by reintroducing *B. longum* into the gut microbiome, it is possible to better regulate these impaired immune systems and relieve the associated symptoms. *B. longum* does this by assisting the immune system in developing a more robust and effective immune response. Based on current understanding, this more robust response can be achieved through a variety of means such as increasing the activity of natural killer cells, reducing the amount of cytokines that have pro-inflammatory effects, or producing more T lymphocytes that are capable of regulating the activity of the immune system (1). While existing studies can characterize the general impacts of *B. longum* on the immune system, the specific mechanisms for certain strains such as *B. longum* BL-10 are poorly understood and need additional research. The aim of this review is to illustrate how the presence of *B. longum* can both regulate and assist in carrying out immune system responses.

Overview of *B. longum* in the Gut Microbiome

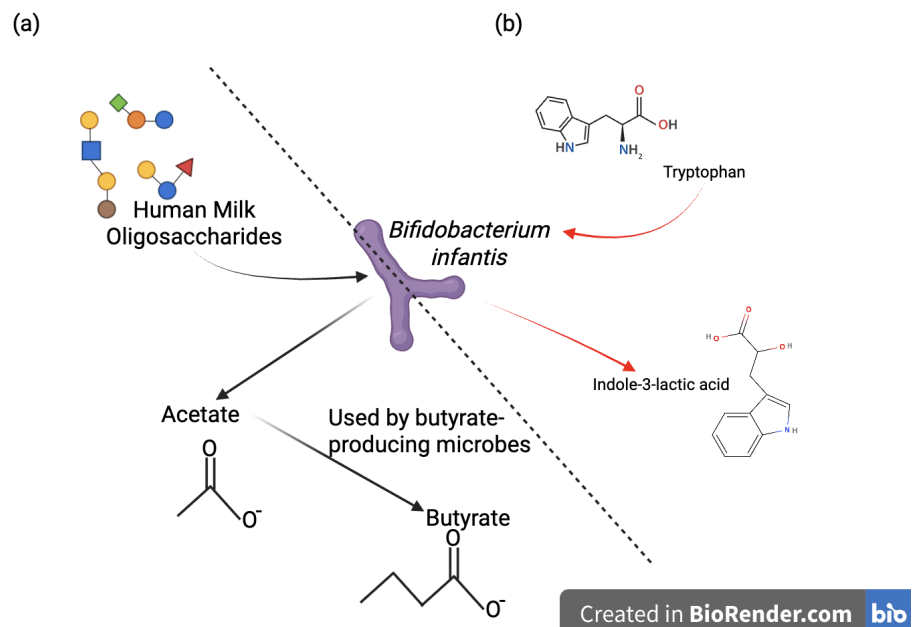
Early on in infancy, one of the microbes that colonizes the gut microbiome is a

subspecies of *B. longum* known as *B. longum* subsp. *infantis* (or simply *B. infantis*). At this stage of life, *B. longum* is one of the most prominent microbes in the gut microbiome. While this microbe will remain in the gut throughout an individual's life, a general trend that has been observed is that the population of *B. longum* gradually declines as they age. Seeing as the many inhabitants of the gut microbiome work in tandem to prevent disease, any changes to this sensitive environment such as microbial imbalances can become problematic. This is the case for individuals who experience gastrointestinal diseases such as irritable bowel syndrome because decreases in the entire *B. longum* population present in the gut have been observed (4). To restore this balance, *B. longum* can be taken as a probiotic supplement as it has been known to correct any damages and replenish the levels of some important metabolites.

While the metabolic capabilities for different strains of *B. longum* will vary, the metabolites that are produced will be influenced by the surrounding environment and modify the immune system. One such metabolite is acetate. *B. infantis* produces acetate as it metabolizes human milk oligosaccharides (or HMOs) in infants (5) and can be seen in figure 1a. The acetate can then be utilized by other microbes in the gut for butyrate production which increases the development of regulatory T cells and regulates the levels of anti-inflammatory cytokines. This was made apparent in observations of lung infections in mice. The mice that had either higher levels of acetate or higher levels of *B. longum* experienced increased amounts of IL-10 (6). Since IL-10 is an anti-inflammatory cytokine, these mice experienced less inflammation during the infection. Another influential metabolite is indole-3-lactic acid. Indole-3-lactic acid is produced when *B. infantis* metabolizes tryptophan and can be used to decrease the production of IL-8 (pro-inflammatory) and can be seen in figure 1b. It can also regulate IL-22 production by interacting with an aryl hydrocarbon receptor that can further decrease inflammation and reinforce the mucosa, preventing the entry of pathogens (6, 7). While acetate and indole-3-lactic acid are just two examples of metabolites that promote gut health and the immune system,

there are many strains and subspecies of *B. longum* that each influence the immune system differently.

Figure 1: (a) An overview of how *B. infantis* metabolizes HMOs into acetate that can then be used to produce butyrate (black arrows). (b) *B. infantis* metabolizes available tryptophan to produce indole-3-lactic acid (red arrows).



The Role of *B. longum* on Early Immune Systems

Of the four subspecies of *B. longum*, the one that is prevalent during infancy is *B. infantis*. This subspecies is commonly found in infants as it primarily utilizes the HMOs that are found in breast milk. By metabolizing HMOs, *B. infantis* produces short chain fatty acids such as acetate and lactate in addition to aromatic lactic acids like indole-3-lactic acid (5, 8, 9). As previously discussed, these metabolites are involved in cellular processes that help regulate the immune system. For instance, short chain fatty acids produced by *B. infantis* like acetate and butyrate can help prevent leaky gut syndrome (5).

As seen in figure 2, the normal intestinal lining is typically impermeable to potential pathogens due to the tight junctions that seal the gaps in between epithelial cells. Additionally, there is mucus that lines many surfaces such as the intestine, trapping and clearing away the

pathogens before they can enter the bloodstream. If an individual has leaky gut syndrome, there are disruptions in the intestinal lining that create openings between the cells. As a result, potential pathogens are able to move past these cells and enter the bloodstream, harming the host. Fortunately, the metabolites made by *B. infantis* help strengthen and reinforce this barrier.

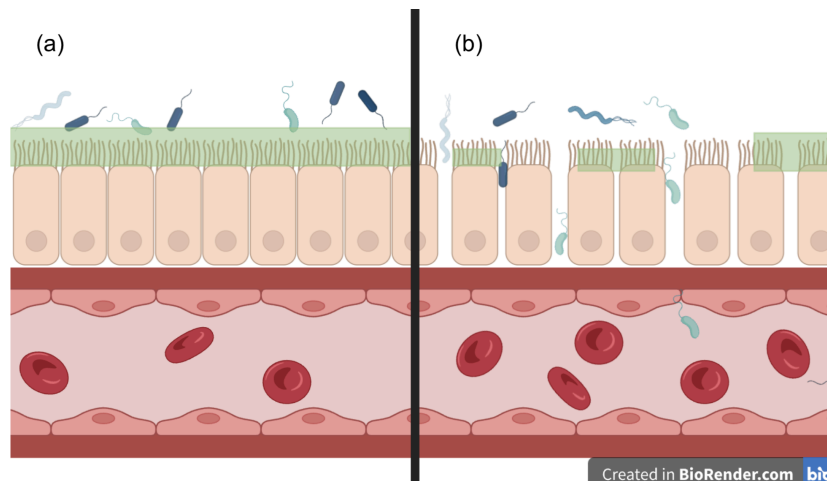


Figure 2: (a) Normal intestinal lining compared to (b) the intestinal lining of an individual with leaky gut syndrome.

Recently, more industrialized countries have transitioned away from breastfeeding and towards formula feeding (8). Since formula feeding often lacks HMOs, populations of *B. infantis* are declining as HMOs are the preferred nutrition source for the microbe. This can eventually lead to the loss of *B. infantis* in the microbiome, hindering the development of the immune system as an individual ages (8).

Impact of *B. longum* on Elderly Immune Systems

While *B. longum* and its subspecies are one of the most prevalent microbes in the gut microbiome early in life, the prevalence of the entire population declines with age. It should be noted that individuals over the age of 100 were a notable exception to this trend as this age group was observed to have larger populations of *B. longum*, specifically the subspecies *longum*, compared to older individuals under 100 years old (10). Generally, as an individual

ages, they experience constant inflammation and a weakened immune system. By reintroducing *B. longum* in older individuals to restore balance in the microbiome, it is possible to regulate this inflammation and the immune system (11). For instance, taking a probiotic supplement that contains *B. longum* leads to decreases in pro-inflammatory cytokines such as TNF- α and greater cytotoxicity of natural killer cells (12). Also, *B. longum* supplementation leads to an increase in anti-inflammatory cytokines such as IL-10 (13). In both studies, elderly individuals experienced less inflammation as a result of balancing the production of these pro- and anti-inflammatory cytokines. Essentially, by restoring the composition of the gut microbiome in older individuals, it is possible to slow down or prevent the functional decline of the immune system as an individual ages.

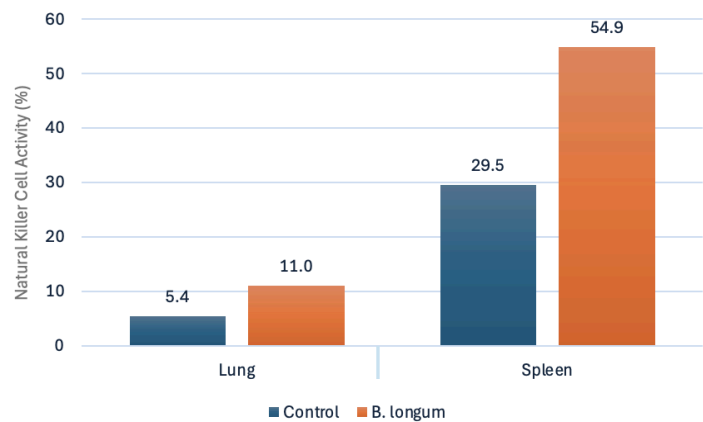
Impact of *B. longum* in Infections

As with many microorganisms naturally present in a host's microbiome, *B. longum* can prevent pathogens from infecting the host. This is largely done through competition for critical resources such as nutrients or competing with the pathogen for space availability. Should a pathogen enter, evade the host's defenses, and activate the immune system, it is necessary to manage the immune response to prevent an overreaction. When mice are exposed to lipopolysaccharides, they experience intestinal damage and inflammation due to elevated levels of pro-inflammatory cytokines and decreased levels of anti-inflammatory cytokines in the serum. On the other hand, when mice are treated with *B. longum* BL-10 and are then exposed to lipopolysaccharides, there were decreases in levels of pro-inflammatory cytokines such as TNF- α and IFN- γ (14). The mice in the control group experienced an unregulated immune response, leading to both intestinal damage and inflammation. Alternatively, the mice treated with *B. longum* had a more controlled response from balancing cytokine levels, minimizing the

amount of intestinal damage. While *B. longum* can influence the immune response by modifying cytokine levels, its presence can also impact the activity of cells and molecules involved in the immune response.

An example of this is seen with the natural killer cells. Mice that were infected with the influenza virus were treated with *B. longum* MM-2. The activity of the natural killer cells found in both the lungs and spleen was then measured (1). As seen in figure 3, there was a significant increase in the activity of the natural killer cells when the mice were treated with *B. longum*. This means that the natural killer cells were able more effectively find and kill the virus infected cells,

Figure 3: The activity (in percent) of natural killer cells found in the lung and the spleen of mice that were (orange) and were not (blue) treated with *B. longum* MM-2 (1).



limiting the spread of the virus. Similarly, to look at the impact that *B. longum* has on the phenotypes of different cells involved in the immune system, a group of males were given a supplement that contained *B. longum* BB536. One noticeable outcome from this experiment was seen in the levels of Foxp3⁺ regulatory T cells. In the first week of taking the supplement, the expression level of this regulatory T cell was 0.45% and, in the third week, the level increased to 1.36% (15). While this increase seems minuscule, it is notable because the expression of this type of regulatory T cells increased three fold. As with natural killer cells, these regulatory T cells help generate a balanced response as they can directly interact with

immune cells or release cytokines that will inhibit the activity of the cell being acted upon. This characteristic can be critical as it prevents immune cells from attacking the host. Another significant component of the immune system is the secretion of antibodies. Immunoglobulin A (IgA) is the antibody most prevalent on mucosal surfaces such as the gastrointestinal lining. Potential pathogens that are found on these surfaces are recognized by IgA, prompting the antibody to bind to the surface of the pathogen. When individuals are exposed to *B. longum* BB536, an observable increase in IgA was recorded. This is best visualized in figure 4 when the concentration of IgA was measured over 16 weeks in individuals who took a placebo and individuals who took *B. longum* BB536 (16). While the concentration of IgA for the placebo group was higher compared to the group that took *B. longum*, when comparing final and initial

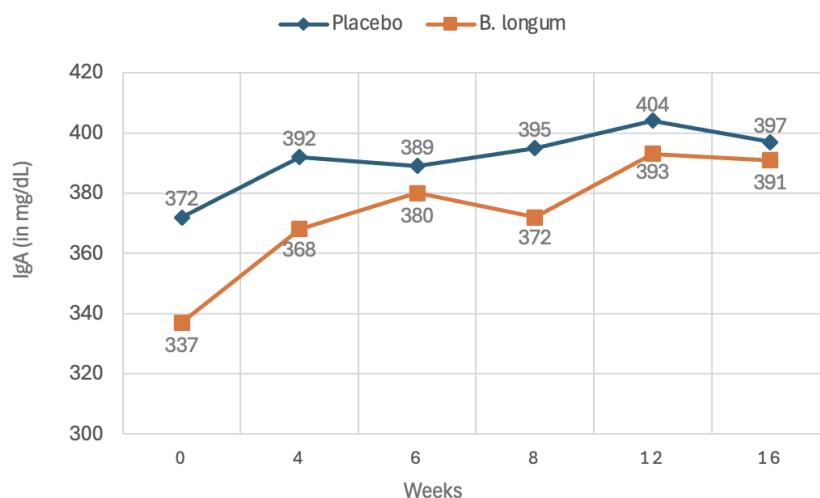


Figure 4: The concentration (in mg/dL) of IgA in both the placebo and treated group over the course of 16 weeks (16).

concentrations for each group, there was a greater difference in concentration for those that took the supplement. This greater increase in antibody concentration is beneficial for the host as antibodies ultimately elicit an immune response such as opsonization, eliminating the pathogen. Evidently, these immunological responses are refined and enhanced in the presence of *B. longum*, allowing for the immune response to occur in appropriate instances.

Role of *B. longum* in Cases of Autoimmunity and Disorders

A response by the immune system is inappropriate when it is unregulated and exaggerated. In an exaggerated response, the cells involved in the immune system interact with substances that are typically not harmful, causing the immune system to react in a way that is damaging to the host. This is the case in autoimmune diseases and disorders such as celiac disease, inflammatory bowel disease, and irritable bowel syndrome.

Individuals with celiac disease experience discomfort and digestive issues when they eat gluten. When consumed, the gluten will be recognized by immune cells, prompting the cells to react in a manner that is similar to the response for pathogens. As a result, this response will cause damage to the villi in the intestine and lead to inflammation. In an attempt to control the immune response, *B. longum* CECT 7347 was used in both animals and humans. In the animal model, rats that had celiac disease were treated with CECT 7347 and had an increase in the production of IL-10 and a decrease in the activation of CD4⁺ T cells (17). As shown in figures 5 and 6, when a human model was used, there were decreases in CD3⁺ T cells, TNF- α , and IFN- γ when *B. longum* CECT 7347 was taken (18). As mentioned previously, regulating the production of cytokines is essential to prevent excessive inflammation and an overactive immune system. In terms of the T cell percentages, a decrease in the percentage means that, due to the decreased signaling and rates of activation in T cells, the immune system itself is less active. In summation, the administration of *B. longum* in subjects with celiac disease led to a greater decrease in inflammatory cytokines and immune cells than that of the untreated group. Both of these trends are significant because a decrease in both means that the immune response that was previously unchecked for individuals with celiac disease is being regulated. Ultimately, this means that the inappropriate immune response involved in celiac disease is being weakened and doing less damage to the host.

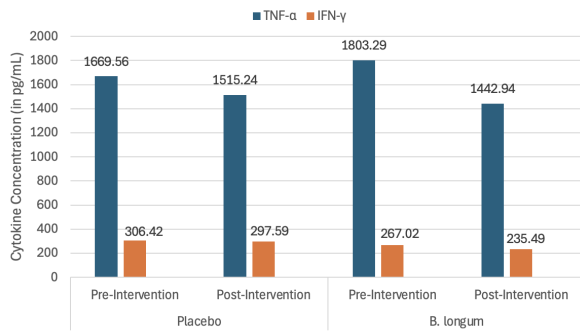


Figure 5: Pre- and post-intervention concentrations (in pg/mL) of TNF- α and IFN- γ for both the placebo and *B. longum* group after three months (18).

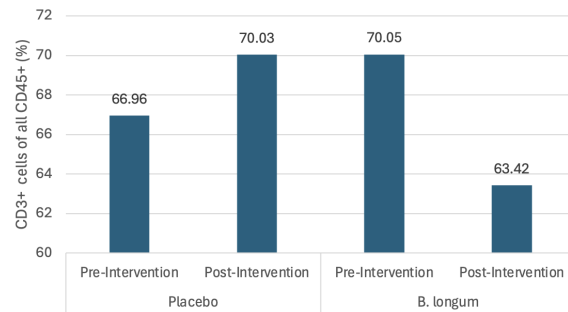


Figure 6: Pre- and post-intervention percentages of CD3⁺ T cells for both groups after three months (18).

As with celiac disease, *B. longum* was also used in an attempt to relieve the symptoms associated with inflammatory bowel disease (IBD). IBD is characterized by excessive inflammation found in the gastrointestinal tract. Yao et al. (2021) notes that diseases such as IBD are generally caused by imbalances in the gut microbiome. More specifically, they explain that the decrease in the entire population of *B. longum* over the course of an individual's life can lead to the development of IBD and be detrimental to components of the immune system such as the gut barrier (19). As seen in other studies, when the population of *B. longum* is restored, there are lower levels of pro-inflammatory cytokines such as TNF- α . Additionally, increasing the general population of *B. longum* was able to reduce the levels of reactive oxygen species through the use of enzymes such as peroxide oxidoreductase and enzymes coded by the *Sir2* gene (19). While reactive oxygen species can be used in the immune system against pathogens, they are dangerous when accumulated in large amounts as they are unstable and reactive. This instability can trigger unwanted reactions, cause damage, and trigger inflammation in the host. By having enzymes that are capable of neutralizing these reactive oxygen species, it is possible to prevent the worsening of inflammation and other IBD symptoms.

Similar to inflammatory bowel disease, irritable bowel syndrome (IBS) can be caused by changes in the gut microbiome, causing discomfort and irregular bowel movements. More specifically, a deficiency in *B. longum* can lead to irritable bowel syndrome. Increasing the population of *B. longum* in subjects with IBS has shown to be useful by enhancing the function of Paneth cells. Normally, Paneth cells contribute to gut health by helping repair any damages to the mucosa and synthesizing antimicrobial peptides that interfere with pathogens. Should Paneth cells not function as they should, the host becomes vulnerable as decreases in antimicrobial peptides and a weakened mucosa impair the immune system. To promote the activity of Paneth cells in rats that have IBS, they were given *B. longum* over the course of ten days. In response to this treatment, Paneth cells were able to resume repairing the mucosa in addition to generating greater amounts of lysozyme (20). Seeing as the Paneth cells of those in the control group were not as efficient at producing antimicrobial peptides such as lysozyme, infections would be more frequent due to the greater number of pathogens bypassing the defenses of the immune system. By increasing the production of lysozyme, pathogens are prevented from forming their peptidoglycan cell wall, causing the bacterium to die. Additionally, repairing the mucosa more often will reinforce the physical barrier that prevents pathogens from spreading throughout the body.

Conclusion

Together, these studies highlight the increasing potential for utilizing *Bifidobacterium longum* to improve the functionality of the immune system. Starting in infancy, *B. longum* subspecies *infantis* has shown to help build the immune system by metabolizing human milk oligosaccharides, preventing leaky gut syndrome. Throughout the course of an individual's life, reintroducing *B. longum* into the gut microbiome can alleviate symptoms and discomfort of

disorders such as celiac disease. Going into old age, *B. longum* was able to improve not only the efficacy of the immune response, but also the regulation of inflammation. While *B. longum* has been shown to improve the functionality of the immune system, further research is required with larger and more diverse sample sizes. Additionally, information concerning the mechanisms of certain strains of *B. longum* remain unknown. For instance, strains such as *B. longum* BL-10 are known to regulate the immune system by influencing both the production of cytokines related to inflammation and influencing surrounding cells. Despite knowing this, some of the exact processes that occur during this mechanism are not known and require future research. This is necessary to have a more complete picture as to how different strains and subspecies of *B. longum* are capable of influencing immune systems.

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